

Qingyun Guo

Curriculum Vitae

Hongze Dist., Huai'an, Jiangsu, China

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Education

- 2026-Now **PhD**, *Electrical Engineering*, Laval University
Canada
- 2022-2024 **Master**, *Control, Robotics and Autonomous Systems*, Aalto University, GPA:4.82/5
Finland
- 2018-2022 **Bachelor**, *Robot Engineering*, China University of Mining and Technology,
GPA:3.73/5
China

Thesis

- Jan. 2024 - **Learning Event-Triggered Algorithms for Networked Control Systems**,
Jun. 2024 Supervisor: Prof. Dominik Baumann
- Customize multi-agent environments by integrating event-triggered mechanisms and modeling network issues.
 - Develop a dynamic communication penalty and extend decentralized policy optimization to hybrid action spaces, enabling joint learning of control and communication policies.
 - Enable selective parameter sharing among homogeneous agents via clustering to improve scalability.
 - Validate the proposed algorithm in both high-dimensional simulation environments and a physical multi-robot system.
- Jan. 2022 - **Design of Amphibious Search and Rescue Robot System for Underground**
Jun. 2022 **Waterlogged Roadways**, *Outstanding graduation design*
- Implement a swing-arm tracked robot design with four auxiliary arms to enhance mobility in waterlogged environments.
 - Conduct dynamic simulations and mechanical design, producing 3D models and technical drawings.
 - Develop a calibration algorithm for the equipped robot arm, enabling motion planning based on the input of the camera and MoveIt.

Publications

- 2026 **Guo, Q.**, Shi, J., Kucner, T. P., Baumann, D. Priority-Driven Control and Communication in Decentralized Multi-Agent Systems via Reinforcement Learning. International Federation of Automatic Control World Congress, 2026.
- 2026 **Guo, Q.**, Shi, J., Huang, J., Shi, T. Individual Control Barrier Functions-Guided Diffusion Model for Safe Offline Multi-Agent Reinforcement Learning. International Federation of Automatic Control World Congress, 2026.

- 2025 Shi, J., **Guo, Q.**, Kucner, T. P. Event-Triggered Maps of Dynamics: A Framework for Modeling Spatial Motion Patterns in Non-Stationary Environments. IEEE/RSJ International Conference on Intelligent Robots and Systems, 2025.

Academic Service

- Reviewer, International Federation of Automatic Control World Congress 2026.
- Reviewer, IEEE/RSJ International Conference on Intelligent Robots and Systems 2025.

Honours and Awards

- Master's degree with honours.
- Finland Scholarship.
- China University of Mining and Technology Scholarship.

Teaching Assistant

- Digital and Optimal Control D: Design homework assignments and lead exercise sessions.
- Basics of Sensor Fusion D: Supervise experiments and guide students through project sessions.

Skills

- Languages: Chinese, English (IELTS: 7).
- Programming Languages: Python, MATLAB, C++.
- Development Tools: VS Code, ROS, CAD, SolidWorks.